



INDRA PUTRA MAHAYUDA

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Madiun, Jawa Timur, Indonesia

I am passionate about Information Technology, with hands-on experience in data science, computer vision, and artificial intelligence, applied in projects involving robotics, electric vehicles, and the mining industry. I have a solid understanding of business process management, data analysis, cloud computing, system integration (API), and database administration. I am detail-oriented, critical in thinking, and capable of working both independently and collaboratively in a team.

Work Experiences

PT Synapsis Sinergi Digital

Aug 2024 - Present

Data Scientist Engineer

- Developed and implemented advanced analytical models to support mining unit operations, focusing on optimizing business processes and improving overall productivity.
- Managed and processed 10M+ data points from diverse systems using cloud-based tools (e.g., BigQuery), performing data cleaning and trend analysis to support operational decision-making.
- Collaborated with cross-functional teams to design and test AI-powered tools, APIs, and dashboards for improving workflow efficiency and system integration.
- Delivered data-driven insights and recommendations to enhance cost-effectiveness and performance in mining operations.
- Technology Stack: Python, SQL, TensorFlow, Google BigQuery, Flask, MLflow, REST API, Git.

PT. VKTR Teknologi Mobilitas, Tbk

Aug 2023 - Feb 2024

Computer Vision Engineer Internship

- Developed radar-camera fusion modules to support early warning features in advanced driver assistance systems, focusing on real-time data integration and processing.
- Implemented object detection and tracking around vehicles using instance segmentation, contributing to intelligent system design and software reliability.
- Worked with embedded systems and real-time communication frameworks to ensure seamless system performance and integration.
- Technology Stack: C, C++, Python, OpenCV, PyTorch, ROS, Linux, TensorRT.

Formulatrix Indonesia (PT. Promanufacture Indonesia)

Feb 2023 - Jul 2023

Software Engineer Internship

- Investigated and resolved camera functionality issues in specialized instrumentation systems, improving hardware- software integration and operational reliability.
- Developed internal tools for camera calibration and contributed to system troubleshooting and automation processes within cross-functional engineering projects.
- Identified and fixed detection system errors, improving accuracy and performance of computer vision pipelines.
- Technology Stack: C, C++, C#, Python, OpenCV, Linux, Git, Jenkins, Raspberry Pi.

EEPIS Robot Soccer (EROS)

Jan 2021 - Aug 2022

Computer Vision Engineer

- Developed an object detection system using deep learning to support visual recognition on a humanoid robot, enhancing system accuracy and responsiveness in real-time environments.
- Built a localization system based on landmark detection to improve navigation and positioning, demonstrating strong skills in software integration and algorithm development.
- Technology Stack: C, C++, Python, OpenCV, TensorFlow, Robot Operating System (ROS), Linux.

Education Level

Politeknik Elektronika Negeri Surabaya

Aug 2020 - Jun 2024

Bachelor Computer Engineering, 3.85/4.00

- Final Project Title : Radar-Camera Fusion for Early Warning System on Advanced Driver Assistance System Application
- Student Activity : Robotics team, UKM Robotics, UKM Software Engineering, UKM Elektrifikasi Transportasi
- Achievements : 3rd Winner Kontes Robot Sepakbola Humanoid division on Kontes Robot Indonesia at National Level (2022), 2nd Winner Kontes Robot Sepakbola Humanoid division on Kontes Robot Indonesia at Regional Level (2022), 1st Winner Kontes Robot Sepakbola Humanoid division on Kontes Robot Indonesia at National Level (2021), 3rd Winner Kontes Robot Sepakbola Humanoid division on Kontes Robot Indonesia at Regional Level (2021), Finalists of CyberSecurity division at KMIPN V (2023), Finalist of Scientific Paper division at GEMASTIK XV (2022).

Organisational Experiences

UKM Elektrifikasi Transportasi

Aug 2023 - Jan 2024

Vice Chairman

- Served as Vice Chairperson of the Student Activity Unit focused on Electrified Transportation, supporting the Chairperson in executing leadership responsibilities and strategic planning.
- Coordinated organizational activities by working closely with division coordinators to ensure smooth operation, effective communication, and successful project execution.

Skills, Certificates & Languages

- **Skills:** Python, C/C++, C#, Machine Learning, Computer Vision, API & Software Development, SQL & Cloud (BigQuery, PostgreSQL, InfluxDB), Linux, System Integration, Data Analysis, Team Leadership, Problem Solving, and Communication.
- **License and Certificate** (2024): BNSP - Junior Web Developer, Huawei Certified ICT Associate - Artificial Intelligence by HUAWEI, Indonesian Young Smart Industry Academy - Talent Scouting Academy by Digital Talent Scholarship, Basic Python Course by Indonesia AI (2021), Belajar Machine Learning untuk Pemula by Dicoding Indonesia (2022), Belajar Dasar Visualisasi Data by Dicoding Indonesia (2022), Memulai Pemrograman dengan Python by Dicoding Indonesia (2022)
- **Language** (2024): Indonesia, English (PEPT TOEFL = 543)